

Problem 2

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Problem 1 Complete steps (i)-(vii) below in order to estimate the following values using linear approximation:

(a) $\cos\left(\frac{31\pi}{180}\right)$

(b) $\sqrt[3]{8.13}$

- (i) Identify the function, $f(x)$.
- (ii) Find the nearby value where the function can be easily calculated, $x = a$.
- (iii) Find $\Delta x = dx$.
- (iv) Find the linear approximation, $L(x)$. [2]
- (v) Compute the approximate value of the expression using the linear approximation.
- (vi) Compare the approximated value to the value given by your calculator.
- (vii) Compare dy and Δy using the value given by your calculator. [2]